

Course Outline

MAD 511 Fundamentals of Android Development 2

****DRAFT******Course ID:** 030557**Academic Year:** 2025/26**Course Description:**

This course is designed to elevate students' proficiency in Android development using Kotlin through advanced tools, architectural patterns, and collaborative software engineering practices. Building on foundational concepts, students will integrate Jetpack components, modern design patterns like MVVM, and persistent storage with Room. Emphasis will be placed on scalable code structure, testing, networking, background processing, and user experience.

		Type:	Not Applicable
Pre-Requisites:	MAD 411	Category:	Vocational
Co-Requisites:	None	Course Credits:	5.00
Special Conditions:	None	Academic Level:	Credit (Post Sec)

Instructional Hours:	Classroom Instruction	30
	Laboratory/Workshops	30
	Other	0
	Total Hours	60

Academic Department:

Windsor:	Zekelman School of Information Technology
Chair:	Ruth Susannah
Chatham:	
Chair:	

Revised By: Mathew Haug**Last Revision:** 2025/05/14

Required Tools, Equipment, and Learning Resources:**Essential Employability Skills (EES):**

Description	Teach	Assess
1) Communication: Communicate clearly, concisely and correctly in the written, spoken, and visual form that fulfills the purpose and meets the needs of the audience.	✓	✓
2) Communication: Respond to written, spoken, or visual messages in a manner that ensures effective communication	✓	✓
3) Numeracy: Execute mathematical operations accurately		
4) Critical Thinking: Apply a systematic approach to solve problems		
5) Critical Thinking: Use a variety of thinking skills to anticipate and solve problems		
6) Information Management: Locate, select, organize, and document information using appropriate technology and information systems	✓	✓
7) Information Management: Analyze, evaluate and apply relevant information from a variety of sources		
8) Interpersonal: Show respect for the diverse opinions, values, belief systems, and contributions of others		
9) Interpersonal: Interact with others in groups or teams in ways that contribute to effective working relationships and the achievement of goals		
10) Personal: Manage the use of time and other resources to complete projects		
11) Personal: Take responsibility for one's own actions, decision and consequences		

Course Learning Outcomes (CLO):

Upon successful completion of this course, the student will be able to:
(EKS = Embedded Knowledge and Skills)

- Design and implement scalable Android applications using advanced architecture patterns.
 (CLO #1)
 EKS:
 - Apply the Model-View-ViewModel (MVVM) pattern to separate concerns and improve testability.
 - Implement repository layers to abstract data sources and promote modularity.
 - Refactor monolithic codebases into modular components for scalability and maintainability

- Utilize Jetpack libraries and components to build responsive, maintainable, and lifecycle-aware applications (CLO #2)
EKS:
 - Use ViewModel to manage UI-related data in a lifecycle-aware way.
 - Integrate LiveData for observable data binding to UI elements
 - Implement Navigation Component with a navigation graph and SafeArgs for seamless transitions.
- Implement secure and efficient networking using Retrofit, coroutines, and modern API consumption practices (CLO #3)
EKS:
 - Perform API calls using Retrofit with coroutine support for non-blocking requests.
 - Parse JSON responses using Moshi or Gson and map them to data models
 - Manage HTTP errors, timeouts, and edge cases with proper error handling strategies
- Apply UI/UX principles and accessibility standards to deliver intuitive and inclusive user experiences (CLO #4)
EKS:
 - Follow Material Design guidelines for consistency and familiarity
 - Use ConstraintLayout and themes/styles to create responsive and scalable UIs.
 - Ensure accessibility compliance using content descriptions, TalkBack testing, and dynamic font support
- Employ background tasks, services, and broadcast receivers to build robust and battery-conscious apps (CLO #5)
EKS:
 - Schedule recurring or delayed tasks using WorkManager with constraints
 - Implement foreground services for persistent background tasks with user visibility
 - Register and respond to system events like battery level or connectivity changes using BroadcastReceiver
- Collaborate within a team using agile practices, version control systems, and code review workflows (CLO #6)
EKS:
 - Use Git/GitHub for version control, branching, and pull requests
 - Apply agile methodologies like sprints, stand-ups, and retrospectives to guide teamwork
 - Conduct code reviews, write meaningful commit messages, and resolve merge conflicts collaboratively
- Demonstrate testing, debugging, and optimization techniques to ensure reliability and performance (CLO #7)
EKS:
 - Write unit tests for business logic and UI tests using Espresso
 - Use Logcat, debugger, and Android Profiler to diagnose and resolve issues.
 - Track and analyze crashes using Firebase Crashlytics or similar tools for post-release

debugging.

- Integrate persistent local storage using Room database and manage asynchronous operations using Kotlin Coroutines (CLO #8)
EKS:
 - Define Room entities, DAOs, and type converters for database modeling
 - Perform asynchronous read/write operations using suspend functions and coroutines
 - Handle background data access with Flow or LiveData to reflect real-time updates in the UI

Teaching/Learning Activities:

Activities may include, but are not limited to:

Active Learning

Discussion (large/small group, online disc. board)

Interactive Lecture

Lecture

Problem Solving/Problem Analysis

Assessment:

Standard/Traditional Delivery

- Assignments 40.00%
 - Frequency:* 4-10
 - Description:* Assignments require problem analysis and the development of programming solutions. Some assignments may require individuals to work in groups. A portion of assignments may be thematically grouped as a project, although each assignment will be graded individually.
 - Outcomes Assessed:* 1, 2, 3, 4
 - EES Assessed:* 1, 2, 6
- Capstone Project 40.00%
 - Frequency:* 1
 - Description:* The students will be required to build an application that clearly demonstrates they are capable of meeting the attached Course Learning Outcomes and their embedded knowledge and skills. The students may be required to also present their application. Group skills will be tracked through the use of online collaboration tools such as GIT and GitHub. Time management will be assessed through milestone and issue tracking.
 - Outcomes Assessed:* 1, 2, 3, 4, 5, 6, 7, 8
 - EES Assessed:* 1, 2, 6
- Group Work 10.00%
 - Frequency:* None
 - Description:* Group work may require individuals to provide feedback to other members within the group. Group work will be assessed through a series of deliverables and tracked through the use of online collaboration tools such as GIT and GitHub, and Issue. Time

management will be assessed through milestone tracking.
Outcomes Assessed: 1, 2, 3, 4, 5, 6, 7, 8
EES Assessed: 1, 2, 6

• Topic Quizzes

10.00%

Frequency: 5-10

Description: Self-test online quizzes used to progressively evaluate the level of understanding of each topic covered in the course.

Outcomes Assessed: 4, 5, 6, 7, 8

EES Assessed: 1, 2, 6

100%

Note: The assessment listed in this outline represents the planned assessment method for this course. Unanticipated conditions during the delivery of the course may necessitate changes to the planned assessment. Students will receive reasonable advance notice should any changes be necessary.

Grading:

A=80-100%

B=70-79%

C=60-69%

F=59% or below

Course Content:

See EKS Statements

"Academic misconduct, including cheating of any form, will not be tolerated. Consequences may include, but are not limited to, a warning, a grade of "0" on the assignment/test/examination, or a failing grade in the course."

(Code of Students Rights and Responsibilities: Section 7.1.6)

All students and employees of this College have a right to study and work in an environment that is free from harassment and discrimination.

Accommodation Statement

The College will provide supports and services to all students with disabilities, both temporary and permanent, with valid supporting documentation. Interim accommodation requests will be received in good faith and can be provided pending receipt of medical documentation. Retroactive accommodations will be considered based on the unique circumstances of the individual matter. The College will give all Human Rights Code-related requests for accommodation meaningful consideration.

Procedure: The student is responsible to meet with a counsellor in Accessibility Services to discuss their functional limitations and accommodation needs and provide Accessibility Services with supporting documentation. Students are not required under the Ontario Human Rights Code to disclose their disability diagnosis (with the exception of Learning Disabilities) to receive accessibility supports and services and/or academic accommodations. Students are encouraged to meet with a counsellor prior to the start of a semester to provide information and arrange accommodations.